8ai	Energy of photon released from A to D = $0 - (-3.14 \times 10^{-19}) = hc/\lambda$ $\lambda = 6.63 \times 10^{-34} \times 3.00 \times 10^{8} / 3.14 \times 10^{-19} = 6.33 \times 10^{-7}$ m = 633 nm 400 nm < λ < 700 nm is the wavelength of visible light range.	M1 A1
aii	By conservation of energy, KE of incident electron = excitation energy (E to A) of cool (ground state) atom + KE of recoiled electron KE of recoiled electron = KE of incident electron - excitation energy (E to A) of cool (ground state) atom KE of recoiled electron = $1.5 \times 10^{-18} - 14.72 \times 10^{-19} = 2.8 \times 10^{-20}$ $\frac{1}{2} m v^2 = 2.8 \times 10^{-20}$ $v^2 = 2 \times 2.8 \times 10^{-20} / 9.11 \times 10^{-19} = 6.147 \times 10^{10}$ $m^2 s^{-2}$ $v = 2.48 \times 10^5$ $m s^{-1}$	C1 C2 A1
aiii	With an incoming electron of 1.44×10^{-18} J, maximum possible energy for the excited electron is $(-14.72 \times 10^{-19} + 1.44 \times 10^{-18}) = -0.32 \times 10^{-19}$ J or level C. (below level B) 3 Possible transitions from level C are C$\rightarrow$ D , C$\rightarrow$ E and D$\rightarrow$ E (-1 for each missing)	B1 B1 B1
aiv	Longest wavelength corresponds to smallest change in energy i.e. C$\rightarrow$ D $hc/\lambda = (-0.49) - (-3.14) \times 10^{-19} = 2.65 \times 10^{-19}$ $\lambda = 6.63 \times 10^{-34} \times 3.00 \times 10^{8} / 2.65 \times 10^{-19} = 7.506 \times 10^{-7}$ m = 751 nm	C1 A1
b	It can be deduced that there are <u>discrete energy levels in the nucleus / the energy levels in the nucleus are quantised.</u>	B1
ci	Hydrogen atoms in the ground state absorb photons from the white light source to go into excited states. When the atoms return to the ground states, photons are emitted <u>in all directions</u> which form the hydrogen emission spectrum. Line emission spectrum of hydrogen is observed.	M1 A1
cii	When the “white” light passes through the cool hydrogen atoms, certain photons which can excite the atoms are absorbed from the incident white light by the ground state hydrogen atoms. The emergent white light have some specific dark lines which correspond to the wavelengths of the absorbed photons. Absorption line spectrum is observed.	M1 A1
di	By conservation of energy, KE gained by electron = EPE lost by “E-field & electron” system $= qV = 1.6 \times 10^{-19} \times 40 \times 10^{3}$ $= 6.4 \times 10^{-15}$	 C1 A1
dii	Minimum wavelength is lower Characteristic spectrum at same position.	B1 B1